

GTFS2GMNS Converter

User's Guide

ASU Trans + AI Lab

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1 Overview

GTFS2GMNS is an open-source Python tool that enables users to conveniently construct any multi-modal transit network in GMNS format from GTFS data. The source code is available at: <https://github.com/asu-trans-ai-lab/GTFS2GMNS>

The `gtfs2gmns.py` script converts General Transit Feed Specification (GTFS) data into General Modeling Network Specification (GMNS) format. This tool processes public transit data (stops, routes, trips, and schedules) and generates a network representation suitable for transportation modeling and analysis.

This converter transforms GTFS transit data into a GMNS-compatible network that includes:

- **Physical nodes:** Transit stops and stations
- **Service nodes:** Route-specific service points
- **Service links:** Connections between stops along transit routes
- **Boarding links:** Connections between physical stations and service nodes
- **Transfer links:** Connections between nearby stops for passenger transfers

1.1 Getting Started: Recommended Learning Path

For students and new users, we recommend the following learning path:

1. **Read this User Guide:** Familiarize yourself with GTFS2GMNS concepts, inputs, and outputs (Sections 1-3)
2. **Use the Interactive Notebook:** Follow the step-by-step `GTFS2GMNS_Verification_Checklist.ipynb` notebook (https://github.com/itsfangtang/gtfs2gmns/blob/main/GTFS2GMNS_Verification_Checklist.ipynb) in Google Colab to:
 - Clone the repository and access the Phoenix GTFS dataset
 - Check input files and verify data structure
 - Run the GTFS2GMNS conversion
 - Examine output files and verify results
 - Complete the verification checklist
3. **Review Examples:** Study the illustrative example (Section 4) and real-world applications (Section 5)
4. **Explore Applications:** Understand how the generated network can be used for various analyses (Section 6)

Note: The verification checklist notebook is designed for Google Colab and uses the Phoenix transit dataset as the primary example. It provides interactive, step-by-step guidance to help you understand and verify the GTFS2GMNS conversion process.

2 Data Flow in GTFS2GMNS Tool

The data flow structure illustrates how various GTFS input files are processed and mapped to generate GMNS output files. Figure 1 illustrates the complete data flow structure of converting GTFS to GMNS, showing the relationships between input GTFS files and output GMNS files, as well as the key field mappings.

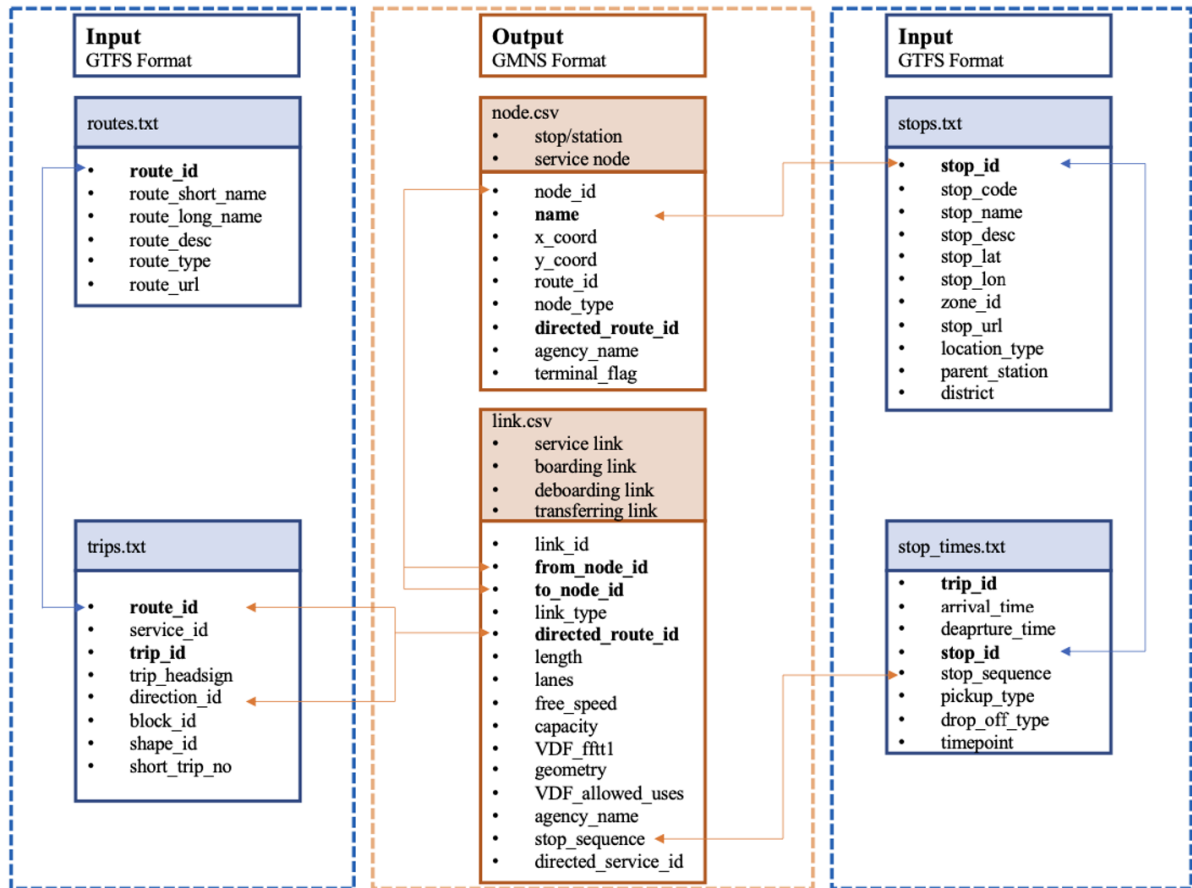


Figure 1: Data flow structure of converting GTFS to GMNS. The diagram shows how GTFS input files (`routes.txt`, `trips.txt`, `stops.txt`, `stop_times.txt`, `agency.txt`) are processed and mapped to generate GMNS output files (`node.csv` and `link.csv`), including the key field relationships and data transformations.

2.1 Input Files (GTFS Format)

The input files of GTFS can be downloaded from GTFS download websites. The required GTFS files include:

- `routes.txt`: Contains route information with fields such as `route_id`, `route_short_name`, `route_long_name`, `route_type`, etc.
- `trips.txt`: Contains trip information with fields such as `route_id`, `trip_id`, `direction_id`, `service_id`, etc.
- `stops.txt`: Contains stop/station information with fields such as `stop_id`, `stop_name`, `stop_lat`, `stop_lon`, etc.
- `stop_times.txt`: Contains stop arrival/departure times with fields such as `trip_id`, `stop_id`, `arrival_time`, `departure_time`, `stop_sequence`, etc.
- `agency.txt`: Contains agency information including `agency_name`

The definition of the above files can be referenced in the official GTFS specification documentation.

2.2 Output Files (GMNS Format)

The conversion process generates a multi-modal transit network in GMNS format, producing the following output files:

- **node.csv**: Contains network nodes with two types:
 - **Physical nodes** (stops/stations): Represent actual transit stops and stations
 - **Service nodes**: Represent route-specific service points

Key fields include: `node_id`, `name`, `x_coord`, `y_coord`, `route_id`, `node_type`, `directed_route_id`, `agency_name`, `terminal_flag`, etc.

- **link.csv**: Contains network links with four types:
 - **Service links**: Connect consecutive stops along transit routes
 - **Boarding links**: Connect physical stations to service nodes (inbound)
 - **Deboarding links**: Connect service nodes to physical stations (outbound)
 - **Transferring links**: Connect nearby physical stations for passenger transfers

Key fields include: `link_id`, `from_node_id`, `to_node_id`, `link_type`, `directed_route_id`, `length`, `lanes`, `free_speed`, `capacity`, `VDF_fftt1`, `geometry`, `VDF_allowed_uses`, `agency_name`, `stop_sequence`, `directed_service_id`, etc.

3 Quick Start Guide

3.1 Option 1: Using Google Colab (Recommended for Students)

The easiest way to get started with GTFS2GMNS is using the interactive Jupyter notebook in Google Colab:

1. Open the `GTFS2GMNS_Verification_Checklist.ipynb` notebook (https://github.com/itsfangtang/gtfs2gmns/blob/main/GTFS2GMNS_Verification_Checklist.ipynb) in Google Colab
2. The notebook will automatically:
 - Clone the repository from GitHub
 - Install required dependencies
 - Load the Phoenix GTFS dataset
 - Configure all necessary parameters
3. Follow the step-by-step instructions in each cell
4. Run each cell sequentially to complete the verification process

The notebook is pre-configured with the Phoenix transit dataset as the example. All paths and parameters are already set, so you can focus on understanding the conversion process rather than configuration.

3.2 Option 2: Local Installation

If you prefer to run GTFS2GMNS locally on your computer, follow the steps in Section 4.

4 Implementation for Your Datasets

To use GTFS2GMNS with your own transit datasets, you need to customize three key parameters in the `__main__` section of the `gtfs2gmns.py` script. Figure 2 shows an example configuration for these parameters and their purposes.

```
if __name__ == '__main__':
    global period_start_time
    global period_end_time
    input_path = './GTFS/BART' ← Input folder of GTFS data
    output_path = './GMNS/BART' ← Output folder
    time_period_id = 1
    time_period = '0700_0800' ← Customized time period
    period_start_time, period_end_time = _hhmm_to_minutes(time_period)

    gtfs2gmns(input_path, output_path)
```

Figure 2: Key parameters to customize for your own datasets. The figure shows the three essential parameters that need to be modified: `input_path` (GTFS data folder), `output_path` (output directory), and `time_period` (time window for trip filtering).

Steps to customize the code for your datasets:

1. **Set the input path:** Modify `input_path` to point to the directory containing your GTFS files. This can be a relative path (e.g., `'./GTFS/YourAgency'`) or an absolute path (e.g., `'/path/to/your/GTFS/data'`).

Option A: Single Agency

```
input_path/
|-- agency.txt
|-- stops.txt
|-- routes.txt
|-- trips.txt
|-- stop_times.txt
|-- [other GTFS files]
```

Option B: Multiple Agencies (Subdirectories)

```
input_path/
|-- Agency1/
|   |-- agency.txt
|   |-- stops.txt
|   |-- routes.txt
|   |-- trips.txt
|   |-- stop_times.txt
|-- Agency2/
|   |-- agency.txt
```

```

|-- stops.txt
|-- ...
|-- ...

```

2. **Set the output path:** Modify `output_path` to specify where you want the GMNS output files (`node.csv` and `link.csv`) to be saved. Ensure this directory exists or will be created by the script.
3. **Set the time period:** Modify `time_period` to filter trips within your desired time window. The format is 'HHMM_HHMM' (e.g., '0700_0800' for 7:00 AM to 8:00 AM, or '1700_1800' for 5:00 PM to 6:00 PM). Only trips that start within this time window will be included in the conversion.

After making these modifications, save the file and run the script. The tool will process your GTFS data and generate the GMNS network files in the specified output directory.

4.1 Verification Checklist

After running GTFS2GMNS, you should verify that:

1. **Output Files Exist:** Check that both `node.csv` and `link.csv` are generated in the output directory
2. **Node Types:** Verify that both physical nodes and service nodes are present in `node.csv`
3. **Link Types:** Verify that all four link types (service, boarding, deboarding, transferring) are present in `link.csv`
4. **Key Fields:** Check that important fields like `VDF_fftt1` (travel time) and `capacity` are populated
5. **Statistics Match:** Compare the number of physical nodes with the number of unique stops in your GTFS data
6. **Data Consistency:** Verify that all link node IDs exist in the node file

For detailed step-by-step verification instructions, refer to the `GTFS2GMNS_Verification_Checklist.ipynb` notebook (https://github.com/itsfangtang/gtfs2gmns/blob/main/GTFS2GMNS_Verification_Checklist.ipynb), which provides interactive code cells to automatically check all these items.

5 Illustrative Example

5.1 Transit Service Network Construction

This section provides an illustrative example demonstrating how GTFS data is transformed into a GMNS network model using a two-layer approach: the **Service Layer** and the **Topological Layer**.

The network model consists of two interconnected layers:

- **Service Layer:** Represents operational transit services with routes S1 (blue) and S2 (red). Service nodes (e.g., 10001, 20005) represent specific stops on routes.
- **Topological Layer:** Represents physical infrastructure with stations (nodes 1-6) and physical connections between them.
- **Interconnections:** Dashed lines connect service nodes to their corresponding physical stations, enabling boarding and alighting.

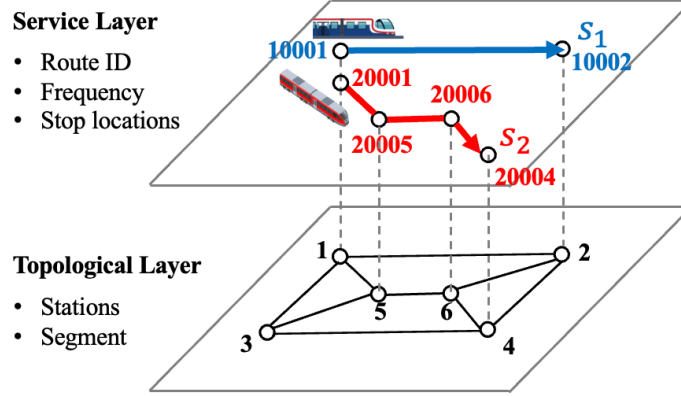


Figure 3: Illustrative Example for Transit Service Network Construction

5.2 Node Information

Table 1 provides examples of different node types in the network:

Table 1: Node Information			
Node	Node Type	Route	Zone
1	Physical node		Zone 1
4	Physical node		Zone 4
10001	Service node	Route S1	
20004	Service node	Route S2	

5.3 Link Information

Table 2 illustrates different types of links in the network:

Table 2: Link Information			
Link	Link Type	Route	Frequency
1 → 10001	Boarding link		
10001 → 1	Boarding link		
10001 → 10002	Service link	Route S1_1	Frequency: 2
20006 → 20004	Service link	Route S2_1	Frequency: 3
20004 → 20006	Service link	Route S2_2	Frequency: 3
2 → 6	Transferring link		

Note: Routes S2_1 and S2_2 represent different segments or directions of route S2.

Link Type Explanations:

- **Boarding Links** (1 → 10001, 10001 → 1): Connect physical stations to service nodes, enabling passengers to board or alight from transit services.
- **Service Links** (10001 → 10002, etc.): Represent segments of transit routes connecting consecutive stops. The frequency indicates the number of trips during the specified time period.
- **Transferring Links** (2 → 6): Connect nearby physical stations, allowing passengers to transfer between different routes or services.

This example demonstrates how the GTFS to GMNS converter transforms raw transit data into a structured network model suitable for transportation modeling and analysis.

6 Real-world Example Application

6.1 BART System as a Test Case and Benchmark

The Bay Area Rapid Transit (BART) system serves as an excellent alternative test case and benchmark for validating GTFS2GMNS functionality. BART is a comprehensive rail transit system serving the San Francisco Bay Area, providing a real-world example of a complex multi-line transit network.

6.1.1 BART System Overview

Figure 4 shows BART operation with multiple lines serving different regions:



Figure 4: BART system overview showing the transit network map in the San Francisco Bay Area. The diagram illustrates the multi-line rail network with 50 physical stations serving San Francisco, Oakland, East Bay, Peninsula, and San Jose regions.

- **Red Line:** Richmond - Millbrae
- **Orange Line:** Richmond - Berryessa/North San José
- **Yellow Line:** Antioch - SFO/Millbrae
- **Green Line:** Berryessa/North San José - Daly City
- **Blue Line:** Daly City - Dublin/Pleasanton
- **Oakland International Airport Line:** Connects Coliseum station to Oakland International Airport

BART GTFS data can be downloaded from the official BART developer resources: <https://www.bart.gov/schedules/developers/gtfs>. The GTFS feed is regularly updated and provides comprehensive schedule and route information for the entire BART system.

6.1.2 Implementation of BART network

To use BART as a benchmark for testing GTFS2GMNS:

1. Download the BART GTFS feed from the official website
2. Extract the GTFS files to a directory (e.g., `./GTFS/BART`)
3. Configure the script with BART-specific parameters:

```
1 input_path = './GTFS/BART'
2 output_path = './GMNS/BART'
3 time_period = '0700_0800' # Morning peak period
```

4. Run the conversion and verify the output statistics match expected values
5. Check that all routes and stations are properly represented in the output files
6. Validate network connectivity by examining service links and transfer links

6.2 Example 1: Single Agency Conversion (BART)

```
1 input_path = './GTFS/BART'
2 output_path = './GMNS/BART'
3 time_period = '0700_0800' # Morning peak: 7:00 AM - 8:00 AM
```

6.3 Example 2: Multiple Agencies Conversion

```
1 input_path = './GTFS/SF' # Contains multiple agency subdirectories
2 output_path = './GMNS/SF'
3 time_period = '1700_1800' # Evening peak: 5:00 PM - 6:00 PM
```

Figure 5 shows the GTFS data of 30 agencies of San Francisco Bay Area.

6.3.1 BART System Statistics

Figure 6 visualizes the BART service network in QGIS software after running GTFS2GMNS on BART data.

When processing BART GTFS data, users should expect the following approximate statistics (may vary by time period and transfer radius):

- **Agency Name:** Bay Area Rapid Transit
- **Number of Physical tops:** 50 stations
- **Number of Service stops:** 136 stops
- **Number of Routes:** 14 routes
- **Number of Trips:** Approximately 2,620 trips (varies by time period)
- **Number of Stop Time Records:** Approximately 38,273 records (varies by time period)

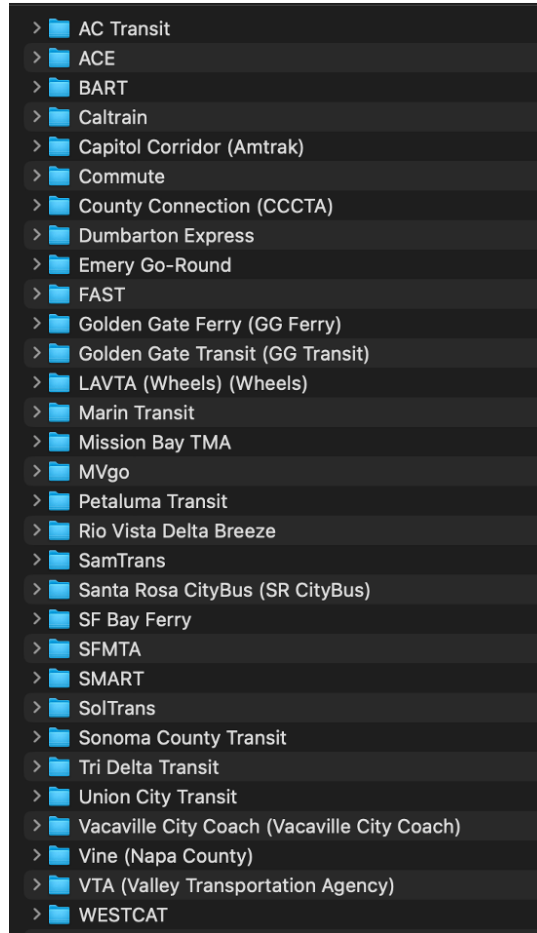


Figure 5: GTFS data of multiple agencies in the folder of "SF"

These statistics can be used to verify that the conversion process is working correctly. After running GTFS2GMNS on BART data, users should check:

- The number of physical nodes in `node.csv` should match the number of unique stops
- The number of service nodes should correspond to unique directed service stops
- The number of service links should reflect the route structure
- All routes should be properly represented in the output files

7 Applications

The GMNS network generated by GTFS2GMNS provides a comprehensive representation of transit systems that can be used for various transportation modeling and analysis applications. This section describes the key applications where the converted network can be utilized.

7.1 Traffic Assignment

The GMNS network output from GTFS2GMNS can be directly used for transit traffic assignment analysis. The network structure includes:

- **Service Links:** Represent actual transit routes with travel times calculated from GTFS schedules, enabling realistic path finding and assignment

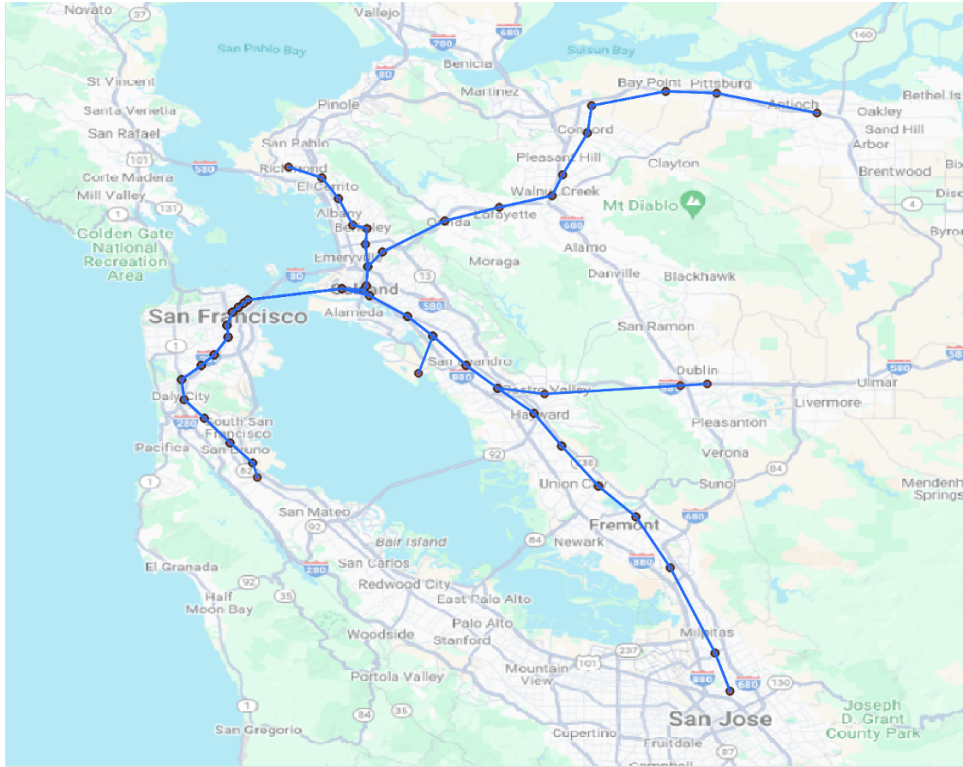


Figure 6: BART service network created by GTFS2GMNS tool

- **Boarding Links:** Model passenger boarding and alighting processes with waiting times based on service frequency
- **Transfer Links:** Enable multi-modal trip assignment by connecting different routes and agencies
- **Link Attributes:** Include capacity, free-flow travel time (VDF_fft1), and volume delay function parameters for congestion modeling

The network can be imported into traffic assignment software (e.g., AequilibraE, MATSim, or custom assignment algorithms) to:

- Assign transit passenger demand to routes based on travel time, transfers, and service frequency
- Analyze route-level and link-level passenger volumes
- Evaluate network performance under different demand scenarios
- Optimize service frequencies and route structures

7.2 Connection Analysis

The GMNS network enables comprehensive connection analysis to evaluate transit system connectivity and service quality:

- **Direct Connections:** Identify direct service connections between origin-destination pairs using service links
- **Transfer Connections:** Analyze transfer opportunities through transferring links, identifying feasible multi-route trips

- **Connection Quality:** Evaluate connection quality based on:
 - Number of transfers required
 - Total travel time (including waiting and transfer times)
 - Service frequency and reliability
 - Transfer penalties based on node types
- **Network Connectivity:** Assess overall network connectivity by examining:
 - Reachability between stations
 - Critical transfer points
 - Isolated or poorly connected areas
 - Redundant connections and alternative paths

This analysis helps transit planners identify service gaps, optimize transfer locations, and improve overall network connectivity.

7.3 Accessibility Calculation

The GMNS network structure is ideal for calculating accessibility metrics, which measure the ease of reaching opportunities (jobs, services, amenities) via transit:

- **Travel Time Accessibility:** Calculate cumulative opportunity measures based on travel time from origins to destinations using the network's travel time attributes (VDF_fftt1)
- **Service Frequency Accessibility:** Incorporate service frequency (represented by lanes in service links) to account for waiting times and service availability
- **Multi-Modal Accessibility:** Leverage the two-layer network structure (physical and service layers) to model:
 - Access to transit from origins (e.g., walking to stations)
 - Egress from transit to destinations
 - Combined transit and non-motorized travel
- **Spatial Analysis:** Use node coordinates (x_coord, y_coord) and geometry fields to:
 - Calculate catchment areas around transit stations
 - Integrate with land use data for opportunity-based accessibility
 - Perform isochrone analysis (areas reachable within specific time thresholds)

Accessibility calculations can support:

- Equity analysis by comparing accessibility across different demographic groups
- Land use and transit planning decisions
- Evaluation of transit investment impacts
- Identification of accessibility gaps and underserved areas

7.4 Equity Analysis

The GMNS network enables comprehensive equity analysis to evaluate how transit services are distributed across different population groups and geographic areas:

- **Spatial Equity:** Analyze the distribution of transit services using node locations and service coverage:
 - Compare service density across neighborhoods with different socioeconomic characteristics
 - Identify transit deserts (areas with limited or no transit access)
 - Evaluate service quality differences (frequency, travel time) across regions
- **Accessibility Equity:** Combine network analysis with demographic data to assess:
 - Accessibility disparities between different income groups, racial/ethnic groups, or age cohorts
 - Job accessibility differences across neighborhoods
 - Healthcare, education, and other essential service accessibility gaps
- **Service Quality Equity:** Evaluate service quality distribution using network attributes:
 - Service frequency (lanes) variation across routes and areas
 - Travel time differences for similar trip purposes
 - Transfer burden (number and penalty of transfers) across different origin-destination pairs
- **Network Structure Equity:** Analyze network topology to identify:
 - Connectivity differences between central and peripheral areas
 - Transfer opportunities and multi-modal access variations
 - Service redundancy and reliability differences

Equity analysis results can inform:

- Transit service planning and resource allocation decisions
- Policy interventions to address accessibility disparities
- Evaluation of transit investments' equity impacts
- Identification of priority areas for service improvements

7.5 Linear Programming Applications

The GMNS network structure from GTFS2GMNS provides an ideal foundation for formulating and solving linear programming (LP) and mixed-integer linear programming (MILP) optimization problems in transit planning. The network's node-link representation and comprehensive attributes enable direct translation into mathematical programming models.

7.5.1 Network Representation for LP Models

The GMNS network can be directly mapped to LP formulations:

- **Nodes as Constraints:** Physical nodes and service nodes can represent:
 - Flow conservation constraints (mass balance equations)
 - Origin and destination nodes for OD pairs
 - Transfer nodes with capacity constraints
- **Links as Decision Variables:** Each link type can represent different decision variables:
 - Service links: Passenger flow variables (x_{ij}) on route segments
 - Boarding links: Boarding flow variables at stations
 - Transfer links: Transfer flow variables between routes
- **Link Attributes as Parameters:** Network attributes provide model parameters:
 - `VDF_fftt1`: Travel time coefficients in objective functions
 - `capacity`: Upper bound constraints on link flows
 - `VDF_penalty1`: Cost coefficients for transfers
 - `lanes`: Service frequency constraints

7.5.2 Common LP Formulations

Transit Assignment Problem The network enables formulation of user equilibrium or system optimal transit assignment. The transit assignment LP minimizes total **travel time** subject to flow conservation and **capacity** constraints:

$$\min \sum_{a \in A} \int_0^{x_a} t_a(\omega) d\omega \quad (1)$$

$$\text{s.t.} \quad \sum_{p \in P_{rs}} f_p = q_{rs} \quad \forall (r, s) \in OD \quad (2)$$

$$x_a = \sum_{p \in P} \delta_{ap} f_p \quad \forall a \in A \quad (3)$$

$$x_a \leq u_a \quad \forall a \in A \quad (4)$$

$$f_p \geq 0 \quad \forall p \in P \quad (5)$$

where:

- A : Set of links (service, boarding, transfer) from GMNS network
- x_a : Flow on link a (decision variable)
- t_a : Travel time on link a (from `VDF_fftt1`)
- u_a : Capacity of link a (from `capacity`)
- f_p : Flow on path p (decision variable)
- q_{rs} : Demand from origin r to destination s
- δ_{ap} : Path-link incidence indicator

Service Frequency Optimization The network structure supports frequency optimization problems:

$$\min \sum_{l \in L} c_l f_l + \sum_{a \in A} w_a x_a \quad (6)$$

$$\text{s.t. } x_a \leq \alpha f_l \quad \forall a \in A_l, l \in L \quad (7)$$

$$\sum_{l \in L} f_l \leq B \quad (8)$$

$$f_l \geq f_{\min} \quad \forall l \in L \quad (9)$$

where:

- f_l : Service frequency on route l (decision variable, related to **lanes**)
- c_l : Operating cost per service on route l
- w_a : Waiting time cost on link a (from boarding links)
- B : Total budget constraint
- A_l : Set of links on route l

Network Design Problem The GMNS network enables transit network design optimization:

$$\min \sum_{a \in A} t_a x_a + \sum_{a \in A} k_a y_a \quad (10)$$

$$\text{s.t. } \sum_{j: (i,j) \in A} x_{ij} - \sum_{j: (j,i) \in A} x_{ji} = b_i \quad \forall i \in N \quad (11)$$

$$x_a \leq M y_a \quad \forall a \in A \quad (12)$$

$$y_a \in \{0, 1\} \quad \forall a \in A \quad (13)$$

$$x_a \geq 0 \quad \forall a \in A \quad (14)$$

where:

- y_a : Binary variable indicating if link a is included in the network
- k_a : Fixed cost of including link a
- b_i : Net supply/demand at node i (from physical nodes)
- M : Large constant

7.5.3 Implementation Workflow

To use the GMNS network for linear programming:

1. **Import Network Data:** Load `node.csv` and `link.csv` into optimization software (Python with PuLP/Gurobi, MATLAB, GAMS, etc.)
2. **Extract Network Elements:**
 - Nodes: Create node sets and identify origins/destinations
 - Links: Create arc sets with attributes (travel time, capacity, cost)

- **OD Pairs:** Define origin-destination demand matrices

3. Formulate Constraints:

- Flow conservation at each node
- Capacity constraints using `capacity` attribute
- Service frequency constraints using `lanes` attribute
- Transfer constraints using transfer link connections

4. Define Objective Function:

- Minimize total travel time (using `VDF_fftt1`)
- Minimize total cost (using `cost` and `VDF_penalty1`)
- Maximize accessibility or service coverage

5. Solve and Analyze: Use LP/MILP solvers to find optimal solutions and analyze results

7.5.4 Advantages of GMNS Network for LP

- **Standardized Format:** GMNS provides a consistent network representation that can be easily imported into various optimization tools
- **Comprehensive Attributes:** All necessary parameters (travel time, capacity, cost) are readily available
- **Two-Layer Structure:** Physical and service layers enable modeling of both infrastructure and operations
- **Multi-Modal Support:** Transfer links facilitate integrated multi-modal optimization
- **Real-World Data:** Based on actual GTFS schedules, ensuring realistic optimization results

7.6 Other Potential Applications

Beyond the applications described above, the GMNS network from GTFS2GMNS can support various other analyses:

- **Network Resilience Analysis:** Evaluate network robustness and identify critical links or nodes
- **Service Planning:** Optimize route structures, frequencies, and schedules
- **Performance Evaluation:** Assess on-time performance, passenger loads, and service efficiency
- **Multi-Modal Integration:** Analyze connections between transit and other transportation modes
- **Scenario Analysis:** Evaluate impacts of service changes, new routes, or infrastructure investments
- **Real-Time Applications:** Support dynamic routing and real-time transit information systems

8 Summary and Next Steps

This user guide has provided a comprehensive overview of the GTFS2GMNS tool, including:

- **Overview:** Understanding what GTFS2GMNS does and the network structure it creates
- **Data Flow:** How GTFS input files are transformed into GMNS output files
- **Quick Start:** Options for getting started (Google Colab notebook or local installation)
- **Implementation:** How to customize the tool for your own datasets
- **Examples:** Illustrative examples and real-world applications (Phoenix and BART)
- **Applications:** Various ways to use the generated network for transportation analysis

8.1 Recommended Next Steps

1. **Complete the Verification Checklist:** Use the `GTFS2GMNS_Verification_Checklist.ipynb` notebook (https://github.com/itsfangtang/gtfs2gmns/blob/main/GTFS2GMNS_Verification_Checklist.ipynb) to:
 - Run GTFS2GMNS on the Phoenix dataset
 - Verify all output files and data structures
 - Answer all checklist questions
 - Understand the relationship between GTFS inputs and GMNS outputs
2. **Explore Your Own Data:** Once comfortable with the Phoenix example:
 - Download GTFS data for your region of interest
 - Modify the three key parameters (`input_path`, `output_path`, `time_period`)
 - Run the conversion and verify results
3. **Apply to Your Research:** Use the generated GMNS network for:
 - Traffic assignment analysis
 - Accessibility calculations
 - Equity analysis
 - Linear programming optimization
 - Other transportation modeling applications

8.2 Getting Help

- **GitHub Repository:** <https://github.com/asu-trans-ai-lab/GTFS2GMNS>
- **Issues:** Report bugs or ask questions on the GitHub Issues page
- **Documentation:** Refer to this user guide and the verification checklist notebook

We hope this guide helps you successfully use GTFS2GMNS for your transportation modeling and analysis needs!